Spectral upper bounds: smeared susceptibilities and moment hierarchies

A nonzero admissible vacuum excitation gives variational upper bounds on the physical Hamiltonian gap. For a gauge-invariant **equal-time spatial** observable, the Feynman–Bijl numerator is its electric derivative norm. At zero momentum, volume cancels against the susceptibility. With a supplied positive spectral measure, consecutive moment ratios give progressively sharper bounds on the observable's spectral edge. That edge can lie above the global gap.

The continuum statements require a finite nonzero Hilbert-space insertion, its spectral representation, ultraviolet control and the stated infrared integrability. Four-dimensional Wilson-flow ultraviolet finiteness does not supply those equal-time or spectral hypotheses: temporal smoothing crosses the reflection plane. The earlier claim that such smoothing eliminated the Hamiltonian assumptions is withdrawn. The exact inequalities, the free spatially flowed Maxwell constants and the gapless inverse-moment countermodel survive under their explicit premises.

Here m denotes an **energy gap**, not a rest mass; a mass is m/c^2. Physical $\hbar$ and c remain explicit. These are written derivations and conditional continuum implications, not a construction of Yang–Mills theory.

Variational bound and intensive susceptibility

Take the temporal-gauge Hamiltonian in the convention of G07 Proposition 7, with g dimensionless and $[A] = L^{-1}$:

$$H = \int d^3x \, \Big[\frac{g^2 c}{2\hbar}\,\Pi_i^a \Pi_i^a + \frac{\hbar c}{4g^2}\,F_{ij}^a F_{ij}^a\Big], \qquad [A_i^a(x), \Pi_j^b(y)] = i\hbar\,\delta^{ab}\delta_{ij}\delta^3(x-y),$$

obtained from $S = -\frac{\hbar}{4g^2}\int F^a_{\mu\nu}F^{a\mu\nu}d^3x\,c\,dt$ with $\Pi_i^a = (\hbar/g^2c)\dot{A}_i^a$.

Proposition 1. Let O be a real functional of the spatial field A alone, gauge invariant, with $\langle O\rangle_0 = 0$ and $O\Omega \in D(H)$. Then

$$E_1 - E_0 \;\le\; \frac{g^2\hbar c}{2}\,\frac{\displaystyle\int d^3x\,\Big\langle \Big|\frac{\delta O}{\delta A_i^a(x)}\Big|^2\Big\rangle_0}{\langle O^2\rangle_0}.$$

Proof. $O\Omega \perp \Omega$, so by min-max $E_1 - E_0 \le \langle O\Omega, (H-E_0)O\Omega\rangle/\|O\Omega\|^2$, and $2\langle\Omega, O(H-E_0)O\Omega\rangle = \langle\Omega,[O,[H,O]]\Omega\rangle$ since $[O,[H,O]] = 2OHO - O^2H - HO^2$. The magnetic term commutes with O. For the electric term, $[\Pi_i^a(x), O] = -i\hbar\,\delta O/\delta A_i^a(x)$ and

$$[O,[\Pi_i^a\Pi_i^a, O]] = -2\big[\Pi_i^a, O\big]^2 = 2\hbar^2\Big|\frac{\delta O}{\delta A_i^a}\Big|^2,$$

so $[O,[H,O]] = \frac{g^2c}{2\hbar}\cdot 2\hbar^2\int d^3x|\delta O/\delta A|^2 = g^2\hbar c\int d^3x|\delta O/\delta A|^2$. Divide by $2\langle O^2\rangle$. □

Only the electric term contributes: the bound measures how much the trial observable is disturbed by the conjugate field, which is the same quantity that Section 2 of the action-floor note identifies as the carrier of $\hbar$.

For a centered observable commuting with H, the numerator vanishes. In a unique vacuum, $O\Omega$ must then vanish as well; a conserved observable cannot give a nonzero centered vacuum fluctuation. The quotient is undefined when that variance is zero. A nonzero zero-energy centered vector would instead diagnose ground-state degeneracy and lies outside the simple-vacuum variational premise.

Zero momentum and volume cancellation

Let $\varphi(x)$ be a gauge-invariant local scalar with $\langle\varphi\rangle_0 = 0$ and take $O = \int_V d^3y\,\varphi(y)$ in a periodic box of volume V. Write

$$D_i^a(x) = \frac{\delta O}{\delta A_i^a(x)} = \int_V d^3y\,\frac{\delta\varphi(y)}{\delta A_i^a(x)}.$$

By translation invariance of the ground state, $\langle|D(x)|^2\rangle_0$ does not depend on x, so the numerator of Proposition 1 is $V\langle|D|^2\rangle_0$, while $\langle O^2\rangle_0 = \int_V\int_V\langle\varphi(y)\varphi(y')\rangle_0^{\mathrm{c}} = V\chi_\varphi$ with $\chi_\varphi = \int d^3z\,\langle\varphi(z)\varphi(0)\rangle_0^{\mathrm{c}}$. The volume cancels:

Corollary 2. For every such φ and every V,

$$\Delta_V \;\le\; \frac{g^2\hbar c}{2}\,\frac{\langle|D_\varphi|^2\rangle_0}{\chi_\varphi}\,,$$

and if $\Delta_V \to m$ and both ground-state quantities converge, the same bound holds for m.

Dimensions: $\delta/\delta A(x)$ carries $L^{-3}/[A] = L^{-2}$, so $[\langle|D|^2\rangle] = [\varphi]^2L^2$ and $[\chi] = [\varphi]^2L^3$; the ratio is an inverse length and $g^2\hbar c$ times it is an energy. The plaquette of the upper-bound note fails precisely because it is a single cell rather than a zero-momentum sum: its variance is an ultraviolet quantity that vanishes in the continuum limit while its gradient does not.

Positive spectral measures and the observable channel

Assume a constructed physical theory with self-adjoint $H \ge 0$, unique vacuum Ω and a specified insertion $v = \Phi\Omega \perp \Omega$, $0 < \|v\| < \infty$. In a finite volume this is an ordinary Hilbert vector; an intensive zero-momentum infinite-volume statement additionally needs the appropriate limit of the spectral measures. For that supplied positive measure,

$$C(\tau) = \langle v, e^{-\tau H/\hbar}v\rangle = \int_{(0,\infty)} e^{-E\tau/\hbar}d\rho(E), \qquad M_k = \int E^k d\rho(E),$$

$$m \le e_\Phi := \inf\operatorname{supp}\rho.$$

The connected insertion removes the vacuum atom. A local observable can miss an entire low-energy sector, so a positive e_Φ alone gives no global lower gap bound.

For $0 < M_k, M_{k+1} < \infty$,

$$m \le e_\Phi \le \frac{M_{k+1}}{M_k}.$$

If $0 < M_{k-1} < \infty$ as well, Cauchy–Schwarz gives

$$M_k^2 \le M_{k-1}M_{k+1}, \qquad \frac{M_k}{M_{k-1}} \le \frac{M_{k+1}}{M_k}.$$

The first inequality uses $E \ge e_\Phi$; the second follows by testing $E^{(k-1)/2}$ and $E^{(k+1)/2}$ in $L^2(d\rho)$. Thus

$$m \le e_\Phi \le \cdots \le \frac{M_{-1}}{M_{-2}} \le \frac{M_0}{M_{-1}} \le \frac{M_1}{M_0}.$$

Each displayed ratio requires its own finite nonzero moments. Tonelli and one-sided differentiation where finite give

$$M_0 = C(0^+), \qquad M_{-1} = \hbar^{-1}\int_0^\infty C(\tau)d\tau, \qquad M_1 = -\hbar C'(0^+).$$

Consequently, **if** $0 < M_{-1} < \infty$ **and** $0 < M_0 < \infty$,

$$m \le \frac{\hbar C(0^+)}{\int_0^\infty C(\tau)d\tau} < \infty.$$

If M_0 is finite but M_{-1} diverges, a positive channel edge is impossible: $E \geq e_\Phi > 0$ would give $M_{-1} \leq M_0/e_\Phi$.

This ratio avoids a derivative numerator **once its spectral and moment hypotheses are supplied**. Ultraviolet finiteness at a point does not prove spatially integrated susceptibility, time-integrated infrared convergence, or admissibility of a four-dimensionally flowed insertion. The spatial flow below stays on one Hamiltonian time slice; its continuum control remains a hypothesis. Sakai–Sasaki, *PRD* **107** (2023) 034510, arXiv:2211.15176v2 (**abstract read**), explicitly distinguish space-time diffusion from spatial gradient flow for spectroscopy. This is source context for that distinction; the abstract is not invoked as a proof of the spectral or continuum assumptions here.

A nonzero finite spectral measure already has some positive weight below a finite energy: $(0, \infty) = \bigcup_n (0, n]$. Thus a nontrivial admissible insertion excludes an empty excited spectrum without requiring M_1 or M_{-1} finite. A useful quantitative bound still needs the corresponding moments. Neither argument establishes $m > 0$.

Spatial flow and the continuum hypothesis

A local operator at a point has divergent gradient in the continuum, so φ must be built from a smeared field. The gauge-covariant smearing is the gradient flow: on the spatial field at fixed time,

$$\partial_s B_i = D_j G_{ji}, \qquad B_i|_{s=0} = A_i, \qquad G_{ij} = \partial_i B_j - \partial_j B_i + [B_i, B_j],$$

the three-dimensional, equal-time counterpart of Lüscher's flow (arXiv:1006.4518v3, equations (1.1)–(1.2), passage level via the local companion), with smearing radius $\sqrt{8t}$ at flow time t and the lattice form given by his equation (1.4). Take $\varphi = \varphi_t = \frac{1}{4} G^a_{ij} G^a_{ij}\big|_{s=t} - \langle \cdot \rangle_0$.

Lüscher's perturbative ultraviolet-finiteness statement concerns the four-dimensional flow in the Euclidean functional integral: expectation values of local gauge-invariant expressions in the flowed field are finite without further renormalization. The object needed here is the equal-time Hamiltonian correlator of the spatially flowed field, which his theorem does not cover. State it as a hypothesis:

> **(F1)** For fixed flow time $t > 0$, the equal-time ground-state susceptibility $\chi_{\varphi_t, a}$ converges to a finite strictly positive limit χ_t, and the complete weighted numerator $g(a)^2 \langle |D_{\varphi_t, a}|^2 \rangle_0$ converges to a finite limit K_t, as $a \to 0$ along the scaling curve (after the specified volume limit).

For the stated nonconstant smooth finite-lattice observable and positive ground state, both quantities are finite and positive and Corollary 2 holds, so (F1) is entirely a statement about the continuum limit. It includes both renormalized Hamiltonian control of the weighted derivative and susceptibility, and nontriviality of the integrated observable. Four-dimensional Euclidean flow does not prove either required equal-time limit by substitution.

Proposition 3 (the finiteness half). Assume the continuum theory exists with a unique vacuum and translation invariance, that the finite-volume gaps converge to m, and (F1) for one flow time t. Then

$$m \;\leq\; \frac{\hbar c}{2} \frac{K_t}{\chi_t} \;<\; \infty.$$

This gives the $m < \infty$ clause conditional on the weighted numerator, the positive susceptibility and spectral-limit hypotheses. The remaining clause, $m > 0$, is untouched: nothing above excludes $m = 0$.

Inverse moments: what finiteness does and does not imply

For a finite measure $M_0 > 0$, a positive channel edge makes every inverse moment finite. The converse fails. For example, with supplied energy $A > 0$,

$$d\rho(E) = \mathbf{1}_{(0,E_*]} e^{-A/E} dE, \qquad E_* > 0,$$

has edge zero and, for $k \geq 2$,

$$M_{-k} = A^{1-k} \int_{A/E_*}^{\infty} u^{k-2} e^{-u} du < \infty.$$

M_0 and M_{-1} are also finite because the upper energy is bounded and the exponential suppresses the lower endpoint. The earlier full gamma expression applies only if the upper integration endpoint is infinite; the truncated countermodel uses this incomplete integral. It has every inverse moment finite while remaining gapless.

Finiteness alone differs from knowledge of the complete sequence. When all inverse moments exist, the ratios M_{-n+1}/M_{-n} tend to e_Φ, possibly zero. One way to see the concentration is to push the finite measure forward by $Y = 1/E$: its L^n norms tend to the essential supremum of Y, and log-convexity makes the consecutive moment ratios tend to the same endpoint. Thus exact full moment data can determine the channel edge; the old assertion that no amount of moment information distinguishes it from a gapped measure was too strong. A global lower gap still needs observable completeness and uniform control over the relevant sectors.

Upper estimates and a gapless example do not close all methods using susceptibility. The complete-frame theorem concerns another operator and has its own lower-bound assumptions. The physical Yang–Mills problem retains the separate spectral-limit obligations.

Free Maxwell inverse-moment checks

For the spatially flowed magnetic scalar computed below, the positive zero-momentum spectral weight is proportional to $k^4 e^{-4tk^2} dk$ at energy $E = 2\hbar ck$. Elementary Gaussian integrals give

$$\frac{M_1}{M_0} = \frac{8}{3\sqrt{\pi}} \frac{\hbar c}{\sqrt{t}}, \qquad \frac{M_0}{M_{-1}} = \frac{3\sqrt{\pi}}{4} \frac{\hbar c}{\sqrt{t}}, \qquad \frac{M_{-1}}{M_{-2}} = \frac{2}{\sqrt{\pi}} \frac{\hbar c}{\sqrt{t}}.$$

The numerical coefficients are approximately 1.504, 1.329 and 1.128. The identities use $\int k^5 e^{-4tk^2} dk = 1/(64t^3)$, $\int k^4 e^{-4tk^2} dk = 3\sqrt{\pi}/(256t^{5/2})$, $\int k^3 e^{-4tk^2} dk = 1/(32t^2)$ and $\int k^2 e^{-4tk^2} dk = \sqrt{\pi}/(32t^{3/2})$, all over $k > 0$. Here M_k exists for $k \geq -4$ and diverges for $k \leq -5$. This power-law example terminates the ratio hierarchy; the exponentially suppressed countermodel above shows that such termination is not necessary for edge zero.

Free Maxwell susceptibility and derivative calculation

The coefficient g^2 in the variational bound is cancelled by the g^{-2} dependence of the ratio of the two free-field observables. Dimensional analysis alone cannot determine this coupling dependence. The following is an exact Gaussian calibration; extending it to an interacting theory near its confinement scale requires nonperturbative control, and an upper bound cannot prove a positive gap.

Free ground state

Take the abelian case of the Hamiltonian above,

$$H = \int d^3x \left[\frac{g^2 c}{2\hbar} \Pi_i \Pi_i + \frac{\hbar c}{2g^2} b_i b_i \right], \qquad b_i = \varepsilon_{ijk} \partial_j A_k, \qquad [A_i(x), \Pi_j(y)] = i\hbar \delta_{ij} \delta^3(x-y),$$

in Coulomb gauge $\partial_i A_i = 0$, two transverse polarizations. In Fourier modes each transverse mode is an oscillator of mass $M = \hbar/(g^2 c)$ and frequency $\omega_k = ck$, since $\frac{1}{2}M\omega_k^2 = \hbar c k^2/(2g^2)$ matches the magnetic term. Its ground-state width gives

$$\langle \hat{A}_i(k)\hat{A}_j(k')\rangle_0 = (2\pi)^3\delta^3(k+k')\,P^{\mathrm{T}}_{ij}(k)\,\frac{\hbar}{2M\omega_k} = (2\pi)^3\delta^3(k+k')\,P^{\mathrm{T}}_{ij}(k)\,\frac{g^2}{2k},$$

with $P^{\mathrm{T}}_{ij} = \delta_{ij} - \hat{k}_i\hat{k}_j$. Dimensions: $[\hat{A}] = L^2$, $[(2\pi)^3\delta^3] = L^3$, and $g^2/2k$ is a length.

Equal-time flow

For an abelian field the spatial flow equation above reduces to $\partial_s B_i = \Delta B_i - \partial_i(\partial_j B_j)$, and a transverse initial field stays transverse, so

$$\hat{B}_i(t,k) = e^{-tk^2}\,\hat{A}_i(k), \qquad \hat{b}_i(t,k) = i\varepsilon_{ijl}k_j\,e^{-tk^2}\,\hat{A}_l(k).$$

The flow time has dimension of length squared; the conventional smearing radius is $\sqrt{8t}$. Set $\varphi_t = \frac{1}{2}(b_t\cdot b_t - \langle b_t\cdot b_t\rangle_0)$, of dimension L^{-4}.

Susceptibility

Write $G_{ij}(z) = \langle b_{t,i}(z)b_{t,j}(0)\rangle_0$. Using $\varepsilon_{iab}\varepsilon_{jcd}k_a k_c P^{\mathrm{T}}_{bd}(k) = k^2 P^{\mathrm{T}}_{ij}(k)$ (the longitudinal part of P^{T} drops because $\varepsilon_{iab}k_a\hat{k}_b = 0$),

$$\tilde{G}_{ij}(k) = k^2 P^{\mathrm{T}}_{ij}(k)\,\frac{g^2}{2k}\,e^{-2tk^2} = \frac{g^2 k}{2}\,e^{-2tk^2}\,P^{\mathrm{T}}_{ij}(k).$$

For a Gaussian field Wick's theorem gives $\langle\varphi_t(z)\varphi_t(0)\rangle_0^{\mathrm{c}} = \frac{1}{2}\sum_{ij}G_{ij}(z)^2$ (two pairings), so by Parseval, with $\sum_{ij}[P^{\mathrm{T}}_{ij}]^2 = \operatorname{tr}P^{\mathrm{T}} = 2$,

$$\chi_{\varphi_t} = \int d^3z\,\langle\varphi_t\varphi_t\rangle^{\mathrm{c}} = \frac{1}{2}\int\frac{d^3k}{(2\pi)^3}\,\frac{g^4k^2}{2}\,e^{-4tk^2} = \frac{g^4}{4}\cdot\frac{1}{2\pi^2}\int_0^\infty dk\,k^4e^{-4tk^2}.$$

With $\int_0^\infty k^4e^{-\alpha k^2}dk = 3\sqrt{\pi}/(8\alpha^{5/2})$ and $\alpha = 4t$, so $(4t)^{5/2} = 32t^{5/2}$,

$$\chi_{\varphi_t} = \frac{g^4}{4}\cdot\frac{1}{2\pi^2}\cdot\frac{3\sqrt{\pi}}{256\,t^{5/2}} = \frac{3g^4}{2048\,\pi^{3/2}\,t^{5/2}}.$$

Dimension: $[\chi] = [\varphi]^2L^3 = L^{-5}$ and $t^{5/2}\sim L^5$, as required.

Gradient term

By Parseval $O = \int d^3y\,\varphi_t = \frac{1}{2}\int\frac{d^3k}{(2\pi)^3}k^2e^{-2tk^2}|\hat{A}(k)|^2 + \text{const}$, so

$$D_i(x) = \frac{\delta O}{\delta A_i(x)} = \big(-\Delta\,e^{2t\Delta}A\big)_i(x) = (\nabla\times b_{2t})_i(x), \qquad \hat{D}_i(k) = k^2e^{-2tk^2}\,\hat{A}_i(k),$$

the curl of the magnetic field at twice the flow time. Hence

$$\langle|D|^2\rangle_0 = \int\frac{d^3k}{(2\pi)^3}k^4e^{-4tk^2}\cdot 2\cdot\frac{g^2}{2k} = g^2\cdot\frac{1}{2\pi^2}\int_0^\infty dk\,k^5e^{-4tk^2} = \frac{g^2}{2\pi^2}\cdot\frac{1}{64\,t^3} = \frac{g^2}{128\,\pi^2\,t^3},$$

using $\int_0^\infty k^5e^{-\alpha k^2}dk = 1/\alpha^3$. Dimension: $[\varphi]^2L^2 = L^{-6}$ and $t^3\sim L^6$, as required.

The bound and coupling cancellation

$$\frac{\langle |D|^2 \rangle}{\chi} = \frac{g^2}{128\pi^2 t^3} \cdot \frac{2048\,\pi^{3/2} t^{5/2}}{3g^4} = \frac{16}{3\,g^2\sqrt{\pi t}},$$

$$m \;\le\; \frac{g^2\hbar c}{2} \cdot \frac{16}{3g^2\sqrt{\pi t}} = \frac{8}{3\sqrt{\pi}}\,\frac{\hbar c}{\sqrt{t}} = \frac{16\sqrt{2}}{3\sqrt{\pi}}\,\frac{\hbar c}{\sqrt{8t}} \approx 4.26\,\frac{\hbar c}{\sqrt{8t}}.$$

Eight independent colour species in the free $SU(3)$ benchmark multiply both numerator and susceptibility by eight, leaving this ratio unchanged. This species cancellation supplies no interacting bound without the admissible insertion, infrared moments and limit assumptions above.

Correction. The earlier version inferred $m \le Cg^2\hbar c/\sqrt{8t}$ from dimensional analysis, assigning the pure numbers k_1, k_2 no coupling dependence. The computation shows $\chi \propto g^4$ while $\langle |D|^2 \rangle \propto g^2$, so the ratio carries g^{-2} and the bound is of order g^0. The corrected statement is that the perturbative bound is $4.26\,\hbar c/\sqrt{8t}\,[1 + O(g^2)]$, with the coupling entering only through the corrections. The t-dependence $1/\sqrt{8t}$ of the earlier note stands.

Two-photon sum-rule check

By the positive spectral representation below, the bound equals the mean excitation energy weighted by the spectral weight of O. In the free theory O is quadratic in $\hat{A}$ and creates two photons of momenta k and $-k$, of total energy $2\hbar ck$, with weight proportional to $\left[k^2 e^{-2tk^2}\right]^2 \left[g^2/2k\right]^2 k^2\,dk \propto k^4 e^{-4tk^2}\,dk$ after the phase-space factor. Therefore

$$\frac{\int_0^\infty 2\hbar ck \cdot k^4 e^{-4tk^2}\,dk}{\int_0^\infty k^4 e^{-4tk^2}\,dk} = 2\hbar c\,\frac{1/(64t^3)}{3\sqrt{\pi}/(256\,t^{5/2})} = 2\hbar c \cdot \frac{4}{3\sqrt{\pi t}} = \frac{8}{3\sqrt{\pi}}\,\frac{\hbar c}{\sqrt{t}},$$

the same number. The agreement checks Proposition 1, the intensive form of Corollary 2 and the Gaussian computation against each other, since both evaluate the same spectral sum rule by different arithmetic routes.

In a weakly coupled regime the free value is a perturbative orientation, not a uniform confinement-scale bound. Its decrease at large flow time is consistent with a gapless photon; monotonic convergence of the interacting spectral mean to the lowest energy is not proved.

Consequence for STATE

The maintained upper-bound family consists of a regulated variational identity, a conditional intensive continuum limit and a positive-measure hierarchy. Spatial smearing, temporal admissibility, infrared integrability, surviving spectral weight and observable completeness retain separate hypotheses. The free calibration and inverse-moment countermodel prove no positive global gap. None of these corrections supplies new research or removes the separate continuum construction and lower-bound obligations.